MP2 — Selecting Reliable LLM Reasoning

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1 Research question and experiment design

Overview and hypothesis

This project requires selecting the candidate most likely to be correct from four supplied solutions to each question. The methodology is to calculate candidates' scores and select the highest-scoring one. Focusing on the research question "Should a solution be judged on its own?", the experiment investigates whether scoring a candidate in the context of the other three leads to more accurate answer selection than scoring each candidate independently. The hypothesis is yes.

For independent scoring, I trained a logistic regression classifier on the supplied 1024-dimensional representation of each candidate's complete solution. For contextual scoring, I kept exploring based on these independent scores. I first introduced a fixed agreement reward based on how many other candidates produce the same final answer and then developed a learned comparison model that combines the independent score with features describing answer-group support, competing candidates, arithmetic contradictions and surface characteristics to produce an overall selection score.

All weights were learned from supplied training data. Out-of-fold first-stage scores were used to train the second-stage classifier. I also found that restricting model fitting to training questions with at least two distinct valid candidate answers improves model overall performance. The final 14-feature selector achieves 291/355 (81.97%) on answer-disagreement validation questions and 385/467 (82.44%) on the complete official test bank. Under the evaluated protocols, candidate comparison improves selection, but repeated incorrect answers can still mislead the selector. Additional text features, geometric relations and a Transformer trained from scratch do not provide reliable incremental gains over the selected model.

Data and evaluation focus

The course provides 4,755 training, 798 validation and 467 test questions. Training/validation mix single-answer groups with genuine answer disagreements while the course test set requires at least two distinct valid answers per question, which is called difficult questions hereafter.

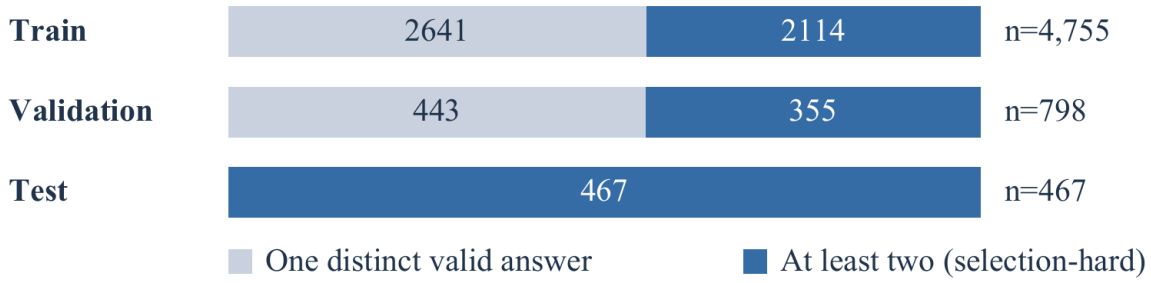


Figure 1. Difficult questions composition

This composition motivated a dedicated analysis of answer-disagreement questions. The distinction describes how much candidate selection is required; it does not label mathematical complexity or identify questions our model gets wrong.

2 Method and feature engineering

Fixed-Weight fusion method

I used logistic regression on each candidate's 1,024-dimensional embedding as the independent-scoring baseline. To incorporate information from other candidates, I first added a majority-voting-based agreement term:

$$score_final_i = score_regression_i + \lambda \times M_i$$

M_i counts the other candidates whose parsed final answers match candidate i's answer. Among all the weights evaluated, λ=0.5 achieved the highest validation accuracy. This weight is fixed across all questions.

Feature engineering

The fixed-weight fusion method captures only the number of candidates sharing the same answer and assigns this agreement term a manually chosen weight. I then extended this approach by constructing 14 features that described

- comparison, to capture answer agreement and the relative strength of supporting and competing candidates;
- arithmetic performance, to indicate whether the candidate contains detectable arithmetic contradictions and how many;
- surface, to describe the candidate's length, formatting, use of digits, and confidence-related wording.

These features are derived from the independent regression scores and parsed answers. A second-stage logistic regression then learns their weights from training data to produce the final candidate score.

Block	Feature	Computation	Interpretation
Comparison	Own_score	Score provided by regression above	Independent candidate quality
	support	v	How many others give the same answer
	Supporter_quality	m - s	Group mean relative to this candidate
	Strongest_rival	c - s	Best competing candidate's advantage
	Group_quality	m - g	Own answer group versus best rival group
	Support_interaction	v(s - c)	Combine support with individual advantage
Arithmetic	Any_contradiction	1 if detected, else 0	At least one supported equality is false
	Contradiction_count	log(1 + count)	Amount of detected arithmetic conflict
Surface	Text_length	log(1 + characters)	Candidate length
	word_count	log(1 + words)	Candidate verbosity
	newline_density	Newlines / characters	Layout and segmentation
	markup_density	Count of *, #, $, backslash / characters	Formatting intensity
	digit_density	Digit characters / characters	Numeric content density
	confident_wording	Confident-word occurrences / words	Frequency of literal confidence cues

Candidates with the same official-parsed final answer form a group. Let v count the other candidates in that group, m be its mean independent score (including self), c be the highest different-answer candidate score, and g be the highest mean among different-answer groups.

Density denominators are at least one. Confident words are c*learly, obviously, certainly, definitely* and *confident*. Log transforms reduce extreme-count influence. The checker evaluates supported numeric equalities with rational arithmetic and rounding tolerance; unsupported expressions are skipped. Contradictions are soft features, not vetoes: flawed work can yield a scorer-correct final answer. Missing competitor differences and invalid-answer support are zero.

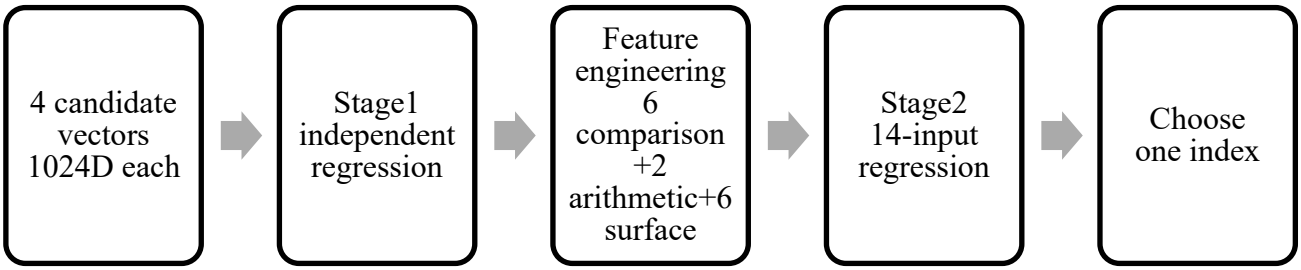


Figure 2. Final inference pipeline

Select the highest-scoring parseable candidate; if none parse, use the highest score.

3 Use difficult questions

We call a question "difficult" when its four candidates contain at least two distinct, non-None answers under the unchanged official numeric parser, compare the questions whose four candidates are all equal mathematically. For example, parsed answers [4, 4, 4, 4] form one group, while [4, 4, 6, 8] form three and qualify as difficult. [4, None, None, None] still has one valid answer and does not qualify. An input audit found that 443/798 validation questions have one distinct valid answer. Even a random choice can solve these questions, which provide very litter information for the model. Those questions measure little ability to distinguish conflicting answers. The remaining 355(798-443) are where candidate comparison matters.

We tested the performance of the same 8-feature comparison-plus-contradiction pipeline fitted on different scopes of training questions.

Training scope	Difficult train - CV (2,114)	Difficult validation (355)	Full validation (798)
All training questions (4,755)	1,654 (78.24%)	283 (79.72%)	726 (90.98%)
Difficult training questions only (2,114)	1,680 (79.47%)	289 (81.41%)	732 (91.73%)
Matched change	**+26 (+1.23%)**	**+6 (+1.69%)**	**+6 (+0.75%)**

Restricting training improves selection on the same difficult evaluation questions. Reduced dilution by easy groups is a plausible explanation.

Surface features are a separate subsequent intervention: with difficult-only training fixed, 8 -> 14 features adds another two difficult validation successes (289 -> 291). The final test result is 385/467 (82.44%), measured only after freezing. We did not run the all-training alternative on test to choose between policies.

4 Training and evaluation protocol

Both classifiers use logistic regression with C=1, max_iter=1000 and random_state=42, preceded by StandardScaler. The final base model fits all four candidates of each difficult training question, including unparseable candidates as negative examples. The meta-model fits only parseable candidates of difficult training questions. This matches the saved frozen experiment.

For training-only estimates, five shuffled outer folds (seed 42) split all 4,755 questions. Within each outer training partition, three inner folds (seed 42) generate held-out base scores for meta-training. The base model is refitted on the outer training partition and the learned meta-model predicts outer-held-out questions. Only input-defined difficult questions contribute to the primary internal metric.

For the final checkpoint, five-fold training out-of-fold base scores generate the final meta-training features. The base classifier then fits all 2,114 difficult training questions. The meta-model is trained on the out-of-fold features, not in-sample final-base predictions. The final meta-model's five-fold cross-fitting differs from the three inner folds used for outer-fold evaluation.

Most lightweight comparisons used the same nested five/three-fold protocol. Scratch-Transformer combinations used a less costly prespecified 80/20 outer holdout (seed 2026) and three-fold final cross-fitting. The selected 14-feature model and source were frozen and hashed before final test evaluation. The portable trainer reproduced every parameter exactly, and portable inference reproduced every frozen validation and test choice.

5 Methods comparison

Training scope	Method	Difficult validation (355)	Full validation (798)
All training	Independent embedding regression	270 (76.06%)	713 (89.35%)

questions (4,755)	Fixed-weight fusion (λ=0.5)	279 (78.59%)	722 (90.48%)
	6-feature selector (comparison)	281 (79.15%)	724 (90.73%)
	8-features selector (comparison+ arithmetic）	283 (79.72%)	726 (90.98%)
Difficult training questions only (2,114)	8-features selector (comparison+ arithmetic）	289 (81.41%)	732 (91.73%)
	14-feature selector (comparison+ arithmetic+ surface)	291 (81.97%)	734 (91.98%)

The first comparison isolates using other candidates while holding the independent scorer and parseability control fixed. Learned group-quality comparisons improve on this independent scorer by 11 difficult validation questions. The additional contradiction block improves by two.

Fixed agreement was less consistent in five repeated 80/20 training holdouts: weight 0.5 changed correct totals relative to the corresponding regression by -7, -1, +5, -4 and -4. The holdouts overlap; these are not five independent test sets. A margin-dependent fusion gate did not beat fixed fusion on validation.

Removing the group-quality and interaction block from the learned selector, leaving only own score and vote count, reduced full validation correctness from 724 to 706. The two-feature classifier learned a large effective vote reward (about 6.30). This ablation supports retaining quality comparisons, although binary candidate training need not optimize question-level selection directly.

Holding difficult-only training and the comparison/contradiction features fixed, six surface features increase internal held-out correctness from 1,680 to 1,689/2,114 and difficult validation correctness from 289 to 291/355. The matched independent regression scores 1,672/2,114 internally and 283/355 on difficult validation; the final selector adds eight difficult validation successes relative to it.

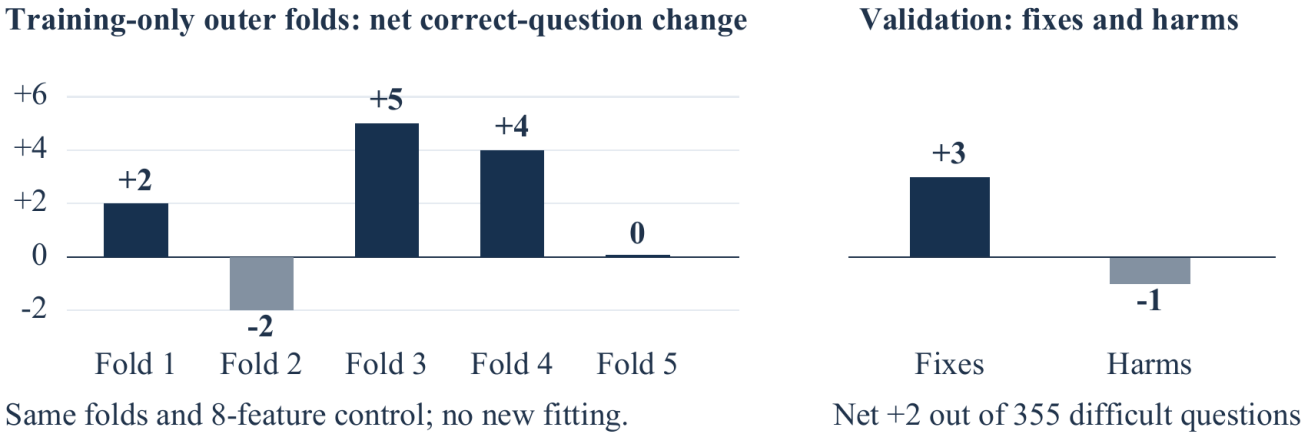


Figure 3. Paired evidence for adding six surface features to the same difficult-trained 8-feature baseline

The internal fold changes are +2, -2, +5, +4 and 0 (19 fixes, 10 harms overall). Validation has three fixes and one harm. The small, mixed result does not establish robustness, significance or a causal benefit of any one surface measurement.

Figure below compares changes against matched baselines on the same 355 difficult validation questions. Additive features retain the baseline; replacements are labeled. Zero net gain does not mean identical choices or absence of useful information.

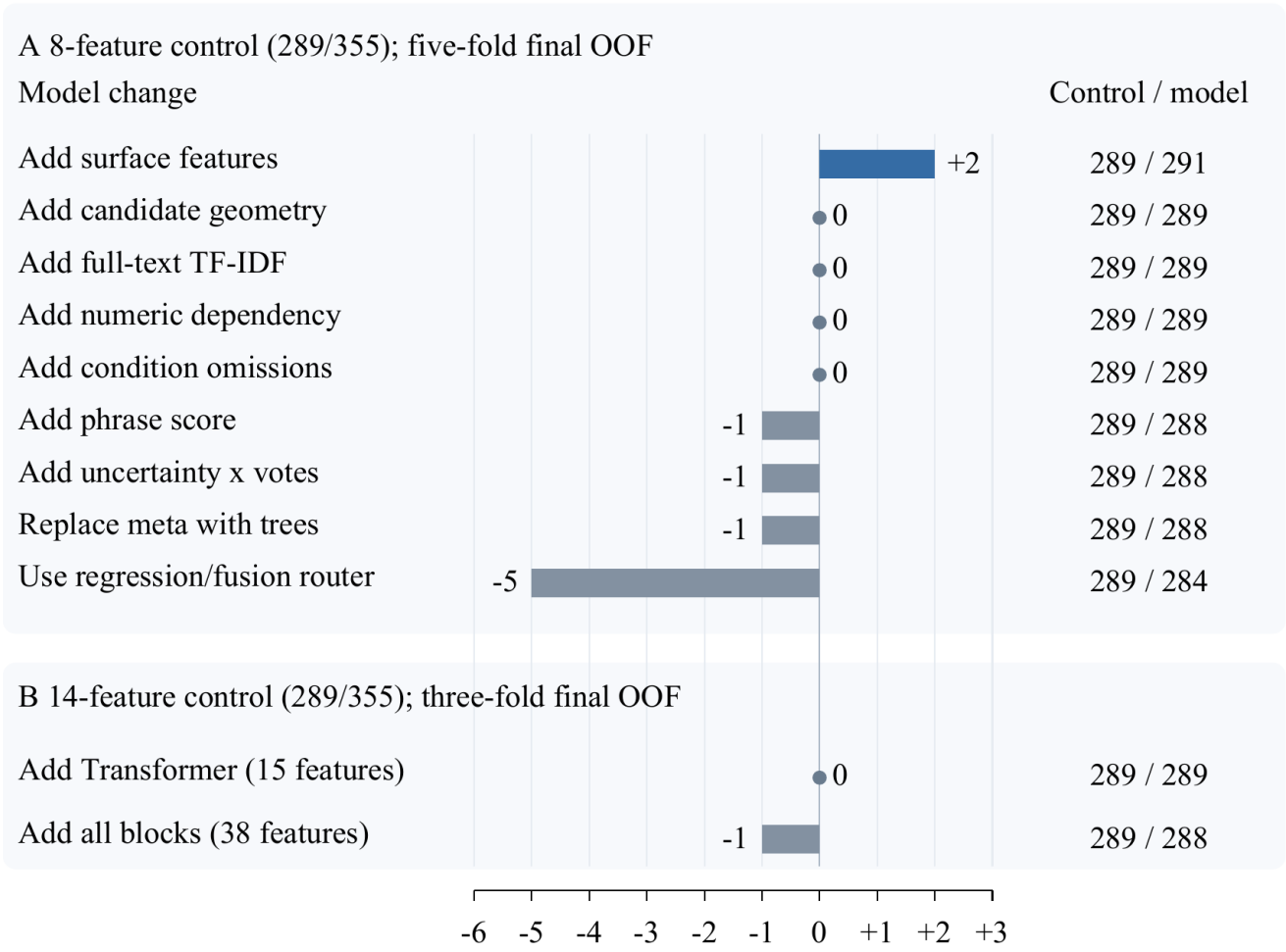


Figure 4. Changes in correctly answered questions

Full-text TF-IDF alone scores 213/355 versus final-answer text 166/355: the text contains predictive information, but this representation is weaker than embeddings. Numeric reuse and literal condition mentions also fall short of semantic verification. The complete experiment matrix remains in MODEL_CATALOG_ZH.md.

The scratch Transformer scores 236/355 versus regression 283/355. Its two-layer architecture has width 64, four heads, FF 128, vocabulary 4000, length 192 and 335,297 parameters; three epochs were selected inside training data. It uses random initialization and final-answer labels, not logic-coherence labels. The 32-unit embedding MLP scores 719/798 versus regression 713/798 but loses all five internal holdouts.

Panel B uses one outer holdout (411 difficult questions) and three-fold final OOF. Its control and 15-feature model score 330/411 internally; 38 features score 328/411. Its 14-feature validation control is 289/355, unlike the selected five-fold model's 291/355. Additional features cannot explain that protocol difference. Neither combination replaces any original 14 features.

A question-level softmax objective did not improve selection. A second training partition supported contradiction-only features (10 fixes, 1 harm) but not equation coverage alone. Together with the vote-only ablation, these controls favor useful evidence over simply increasing complexity.

6 Failure analysis, limitations and disclosure

Wrong agreement remains a limitation

Among the 355 difficult validation questions, 60 have a unique highest-vote answer that is wrong; this includes 2+1+1 plurality cases. On these questions difficult-only independent regression gets 30 correct, while both the 8-feature baseline and selected 14-feature model get 26. Comparing candidates helps aggregate selection but does not solve wrong-majority interference. These gold-defined diagnostic categories are never available to inference routing.

Only 20 validation questions have four different parseable answers. The final surface model gets 12 correct. This small group cannot support strong generalization claims.

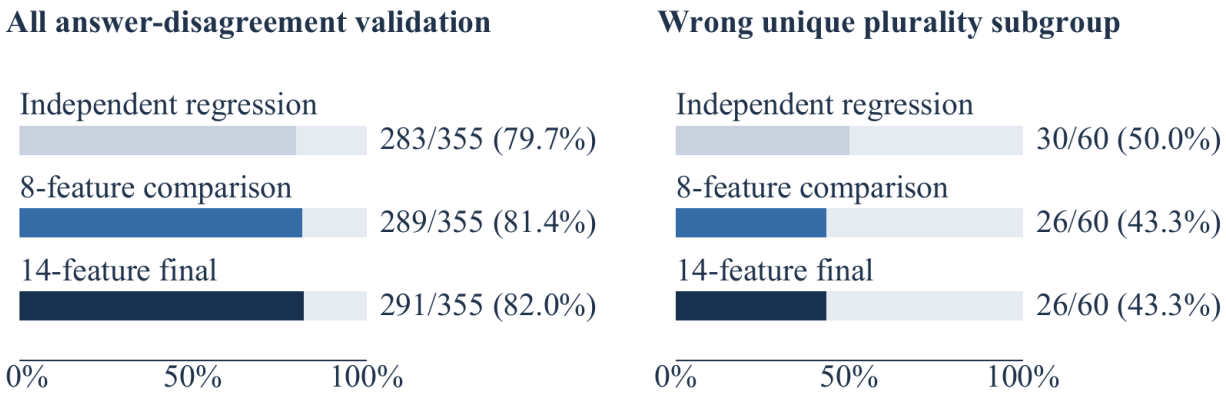


Figure 5. A subgroup tradeoff under identical difficult-only training

Arithmetic correctness is not task correctness

The fixed checker flags contradictions in 195 of 19,020 training candidates. Of those, 166 have wrong final answers and 29 have correct final answers. Its coverage is sparse: it flags only 3.56% of wrong candidates. A numeric-reuse diagnostic also finds 10 training candidates with a correct final answer despite a contradiction whose explicit right side equals that final answer; overlapping ancestry flags are not additive counts. These observations justify soft features instead of a hard arithmetic veto.

A solution may mention every given number and compute each multiplication correctly while assigning a number to the wrong quantity or ignoring a condition. Literal mention coverage, cosine proximity and confidence wording do not establish semantic correctness. The final model is an empirical answer selector, not a proof checker.

Final score and limits of the evidence

The complete official test result is 385/467 = 82.44111349036403%, with 82 errors and no excluded questions. Its proximity to difficult validation (81.97%) is reassuring for this bank, not a guarantee of audit-bank performance. Repeated validation inspection, limited training diversity, small subgroups, correlated candidate errors and the weaker single-holdout Transformer comparisons constrain generalization claims. No post-test optimization was performed.

AI disclosure

Generative AI assisted experimental design, implementation and drafting. I am responsible for reviewing and understanding the final work.